

2. Top 10 Feature Shortlist:

Rank	Clinical Feature	Category	Rationale for Inclusion (Model Utility)
1	esi	Target Variable	The outcome variable defining acuity (1-5 scale). Must have zero missingness.
2	triage_vital_o2	Vitals (Sensor)	Strongest individual separator of ESI classes; distinct signals between stable and unstable cohorts.
3	cc_chestpain	Symptom (Binary)	Direct predictor of cardiac risk; highly sensitive signal for higher-acuity (ESI 2) tracking.
4	triage_vital_hr	Vitals (Sensor)	Essential physiological metric for baseline stability and cardiac output assessment.
5	triage_glucose	Vitals (Lab)	Crucial indicator for diabetic emergencies and metabolic stability (e.g., DKA).
6	age	Demographic	Validated demographic vector with influence on baseline operational resource utilization.
7	triage_vital_temp	Vitals (Sensor)	Key vector for infectious disease monitoring (fever vs. hypothermia indicators).
8	triage_vital_sbp	Vitals (Sensor)	Primary cardiovascular metric (Systolic BP) to define hemodynamic stability.
9	triage_vital_dbp	Vitals (Sensor)	Secondary cardiovascular metric (Diastolic BP) used for validation against SBP.
10	gender	Demographic	Operational vector to be included for model context; must be audited for predictive equity.